Kacper Saks

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SUMMARY

PROCESS & MANUFACTURING ENGINEER (AEROSPACE) · APPLIED AI/ML — MODELLING, VERIFICATION & VALIDATION

Aerospace process engineer at Airbus Defence and Space with an independent applied-AI/ML research track centred on verification and validation. Three years developing technological processes, assembly sequences and control documentation for prototype and serial aircraft production; in parallel, building probabilistic models and reproducible validation pipelines — deflated performance metrics, combinatorial purged cross-validation, calibration and uncertainty quantification — and publishing the results (a self-archived preprint; an open-source research artifact with a Zenodo DOI). FEM-certified; CATIA V5, SAP ERP and Python in daily use.

EXPERIENCE

Sep 2023 — Present

PROCESS ENGINEER

Airbus Defence and Space

Warsaw, Poland

- Progression: Intern (Sep 2023) → Junior Process Engineer (Jan 2024) → Process Engineer (May 2026)
- Develop technological processes, assembly sequences and control cards for parts and subassemblies in prototype and serial aircraft production (Airbus Poland and CASA/Airbus cooperative programmes)
- Optimised production workflows and material/fastener allocation — up to 30% efficiency improvement and ~€50K annual cost savings
- Verify processes against new technical developments, inspection-monitor workstation compliance, and implement design-office changes
- Define tooling, fixture and metrology assumptions; select tools on techno-economic criteria; normalise working time
- Deliver internal training on design & technological documentation; apply Lean standardisation and FOD discipline daily (CATIA V5, SAP ERP, GD&T)

2024 — Present

CO-FOUNDER & CTO

K&S Venture Group — Clashpoint.me

Warsaw, Poland

- Designed, built and operate an AI-driven sales-coaching platform end to end — FastAPI, PostgreSQL, Next.js/React, LLM features

RESEARCH & PUBLICATIONS

> **The Validation Crisis in AI Forecasting** (https://kacpersaks.dev/research/validation-crisis)
> SELF-ARCHIVED PREPRINT · 2026
>
> Deflated Sharpe ratio, a sequential adaptation of PBO (probability of forecast overfitting) and walk-forward retrodiction applied to published AGI capability forecasts — a pre-registered pipeline that reports its own failed prediction.

worldclass-research-org — 39-role autonomous research organisation

(https://kacpersaks.dev/research/executable-gate-research-org)

OPEN SOURCE · ZENODO DOI 10.5281/ZENODO.20645678 (V0.2.0)

Multi-agent pipeline enforcing pre-registration and reproducibility across the research lifecycle (github.com/Ricko12vPL/worldclass-research-org); centrepiece pre-registered gate-calibration study, with a public erratum (v0.2.1) downgrading its cross-arm results to exploratory.

X_Quant — solo quantitative research program (https://kacpersaks.dev/projects/x-quant)

SOLO RESEARCH · 2024-2026

v1 (≈274K LOC) was retired in Aug 2026; its published metrics are withdrawn. v2 rebuilt the process under a written research constitution — pre-registration, an append-only decision register, a permanent attempts ledger and separate build/verify sessions. No strategy it tested was validated for trading; none reached paper or live trading.

Alpha Lab — local research workstation for alpha research

(https://kacpersaks.dev/projects/alpha-lab)

PRIVATE · IN PROGRESS · 2026

Successor to X_Quant: point-in-time data, a fast vectorised backtest engine, a signal lab and a research-integrity engine (hash-chained trial ledger, sealed holdouts, DSR / PBO / SPA). Key phases close with an independent AI-agent review, and the integrity statistics are mutation-tested; no real-return results yet.

neo-triage — hierarchical Bayesian classification (https://github.com/Ricko12vPL/neo-triage)

MANUSCRIPT IN PREPARATION

NumPyro hierarchical Bayesian classifier with a LightGBM baseline for near-Earth-object triage; calibration, uncertainty quantification and a reproducible training pipeline.

SKILLS

VERIFICATION & VALIDATION

Combinatorial Purged CV
Deflated Metrics (DSR) PBO / CSCV
Walk-Forward Calibration & UQ
Pre-registration
Reproducible Pipelines
SPC / Control Cards

AI / ML

Bayesian Modelling (NumPyro)
LightGBM / Gradient Boosting
scikit-learn PyTorch SHAP
LLM & Multi-agent Pipelines RAG

MODELLING & SIMULATION

FEM — Linear & Nonlinear (certified)
Convergence / Mesh-independence
CATIA V5 Autodesk Inventor
Python numerical stack

MANUFACTURING ENGINEERING

Technological Process Design
Assembly Sequencing
Tooling & Metrology GD&T SAP ERP
Lean / Continuous Improvement FOD

BACKEND & PROGRAMMING

Python SQL
TypeScript / JavaScript FastAPI
PostgreSQL Next.js / React Docker
Git

EDUCATION

- Oct 2024 — Aug 2026

  Master's programme — Project & Process Management

  Collegium Civitas

  Warsaw

- Project-based

  SOFTWARE ENGINEERING — 42 WARSAW

  Peer-to-peer coding program

  Warsaw

  — Project-based, peer-to-peer software engineering — C, algorithms, Unix systems

- Oct 2018 — Feb 2022

  BEng Mechanical Engineering

  Silesian University of Technology

  Gliwice

  — Thesis: feed mechanism of mining rigs and bolting machines · Major GPA 4.17 / 5.00

CERTIFICATIONS

FEA Convergence & Mesh Independence Coursera

FEM — Linear, Nonlinear Analysis & Post-Processing Coursera

Certified User: Inventor Autodesk

Introduction to Orbital Mechanics Udemy

Interplanetary Spacecraft & Satellite Engineering Udemy

Google Data Analytics Google

Python for Everybody Univ. of Michigan

LANGUAGES & ELIGIBILITY

Polish (native) · English (professional working proficiency — daily working language at Airbus). Polish national (EU / ESA Member State). Holder of a Polish national personnel security clearance; experienced with export-control and classified-information regimes.